


Matrix factorizations

1. SVD $\longrightarrow$
2. Eigen decomposition
3. QR decomposition
4. PCA
5. LU decomposition
6. ALS
7. iALS

truncated:

$$R \approx U \Sigma V^T$$

$$k \ll \ell = \text{rank}(R)$$

$$U^T U = V^T V = I_k$$

$$r_{ij} = \bar{u}_i^T \Sigma \bar{v}_j$$

$$\min_{X, Y} \sum_{i,j} (r_{ij} - x_i^T y_j)^2 + \lambda \sum_i \|x_i\|^2 \underbrace{C_i}_{\text{about popularity}} + \lambda \sum_j \|y_j\|^2 \underbrace{C_j}_{\text{about popularity}}$$

about popularity

ALS (Alternating Least Squares)

$$\min_{X, Y} \sum_{(i,j) \in R} (r_{ij} - x_i^T y_j)^2 + \lambda \sum_i \|x_i\|^2 C_i + \lambda \sum_j \|y_j\|^2 C_j$$

X - user matrix, Y - item matrix

R

1	x	x	x	x	0
x	x	x	x	x	1
x	1	x	x	x	x
1	x	x	x	x	x
x	0	x	x	x	x

undefined

$$R \approx XY$$

$$R \in \mathbb{R}^{n \times k}, X \in \mathbb{R}^{n \times l}, Y \in \mathbb{R}^{l \times k}$$

$$l - \text{embedding dimension} \quad \frac{\partial}{\partial x_i} (x_i \cdot x_i) = 2x_i$$

$$\begin{aligned} \argmin_{x_i} P &= \argmin_{x_i} \left[\sum_{(i,j) \in R} r_{ij}^2 - 2 \sum_{(i,j) \in R} r_{ij} x_i^T y_j + \sum_{i,j} (x_i^T y_j)^2 + \lambda \sum_i (x_i, x_i) C_i + \lambda \sum_j (y_j, y_j) C_j \right] = \\ &= \argmin_{x_i} \left[-2 x_i^T \sum_{(i,j) \in R} r_{ij} y_j + \sum_i x_i^T y_j y_j^T x_i + \lambda C_i x_i^T x_i \right] = \\ &= \argmin_{x_i} \left[-2 x_i^T \underbrace{\left(\sum_{(i,j) \in R} r_{ij} y_j \right)}_{B_i} + x_i^T \underbrace{\left(\sum_j y_j y_j^T \lambda C_j \right)}_{A_i} x_i \right] \end{aligned}$$

$$\Rightarrow \argmin (-2 x_i^T B_i + x_i^T A_i x_i)$$

$$\frac{\partial}{\partial x_i} (-2 x_i^T B_i + x_i^T A_i x_i) = 0$$

$$\frac{\partial}{\partial x_i} (A x_i^2) = 2 A x_i$$

$$\begin{aligned} \frac{\partial}{\partial x_i} (-2 x_i^T B_i) &= -2 B_i \\ \frac{\partial}{\partial x_i} (x_i^T A_i x_i) &= 2 A_i x_i \end{aligned}$$

$$-2 B_i + 2 A_i x_i = 0 \quad x_i = A^{-1} B$$

iALS (implicit ALS)

implicit ALS $\neq$ implicit feedback

$$p_{ui} = \begin{cases} 1, & r_{ui} > 0 \\ 0, & r_{ui} \leq 0 \text{ or undefined} \end{cases}$$

p_{ui} - preference

$$c_{ui} = 1 + \alpha |r_{ui}|, \quad \alpha \text{ is high here! } (\sim 10^5)$$

c_{ui} - confidence

$$\operatorname{argmin}_{x_i} \sum_{ij} c_{ij} (p_{ij} - x_i^T y_j)^2 + \lambda \sum_i \|x_i\|^2 \hat{C}_i + \lambda \sum_j \|y_j\|^2 \hat{C}_j =$$

? Answer?
$$\left(\sum_{ij} c_{ij} y_j y_j^T + \lambda \hat{C}_i I \right)^{-1} \left(\sum_{ij} c_{ij} p_{ij} y_j \right) =$$

$$= \left(\sum_{j|p_{ij}=0} \overset{=1}{\textcircled{c_{ij}}} y_j y_j^T + \sum_{j|p_{ij} \neq 0} c_{ij} y_j y_j^T + \lambda \hat{C}_i I \right) \times$$

$$\times \left(\sum_{j|p_{ij} \neq 0} c_{ij} p_{ij} y_j \right) =$$

$$= \left(\sum_{j|p_{ij}=0} y_j y_j^T + \sum_{j|p_{ij} \neq 0} c_{ij} y_j y_j^T + \lambda \hat{C}_i I \right)^{-1} \left(\sum_{j|p_{ij} \neq 0} c_{ij} p_{ij} y_j \right) \ominus$$

$$\hookrightarrow \sum_j y_j y_j^T - \sum_{j|p_{ij} \neq 0} y_j y_j^T$$

$$\Leftrightarrow \left(\underbrace{\sum_j y_j y_j^T}_{-1} - \underbrace{\sum_{j|p_{ij} \neq 0} y_j y_j^T} + \underbrace{\sum_{j|p_{ij} \neq 0} c_{ij} y_j y_j^T} + \lambda \hat{C}_i I \right) \left(\sum_{j|p_{ij} \neq 0} c_{ij} p_{ij} y_j \right) \Leftrightarrow$$

$$Y \in \mathbb{R}^{l \times k}$$

$$\text{row} \times \text{col} = \text{number}$$

$$y_j \text{ col}$$

$$\text{col} \times \text{row} = \text{matrix}$$

$$x_i \text{ row}$$

$$X \text{ with row } x_i$$

$$Y \text{ with col } y_j$$

$$\Leftrightarrow \left(Y^T Y + \lambda \hat{C}_i I + \sum_{j|p_{ij} \neq 0} (c_{ij} - 1) y_j y_j^T \right)^{-1} \left(\sum_{j|p_{ij} \neq 0} c_{ij} p_{ij} y_j \right)$$